

5.3 Inverses, Cramer's Rule, Cross Products, and Volume

Concise exam revision notes for adjugates, determinant geometry, and orientation.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Determinants connect algebra and geometry: they decide invertibility, give formulas for inverse entries, and measure signed area or volume.

For an invertible square matrix,

$$A^{-1} = \frac{1}{\det A} C^T,$$

where C is the cofactor matrix. Equivalently,

$$(\det A)I = AC^T.$$

2 Cramer's Rule

For

$$Ax = b,$$

each variable is a determinant ratio:

$$x_i = \frac{\det A_i(b)}{\det A},$$

where $A_i(b)$ is A with column i replaced by b .

Use Cramer's rule when	Avoid it when
The matrix is small or symbolic. You need a formula for one variable.	The matrix is large and elimination is easier. You need all variables numerically.

Common mistake. Cramer's rule requires $\det A \neq 0$. If $\det A = 0$, the determinant ratios do not apply.

3 Area from Determinants

For two vectors in the plane,

$$u = \begin{bmatrix} a \\ b \end{bmatrix}, \quad v = \begin{bmatrix} c \\ d \end{bmatrix},$$

the signed area of the parallelogram is

$$\det \begin{bmatrix} a & c \\ b & d \end{bmatrix} = ad - bc.$$

The absolute value gives the geometric area:

$$\text{area} = |ad - bc|.$$

4 Cross Product

In $\mathbb{R}^3$, the cross product is perpendicular to both input vectors:

$$u \times v \perp u, \quad u \times v \perp v.$$

Using unit vectors,

$$u \times v = \det \begin{bmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}.$$

Its length is the parallelogram area:

$$\|u \times v\| = \|u\| \|v\| \sin \theta.$$

Exam use. Use cross product when a problem asks for a normal vector to a plane in $\mathbb{R}^3$.

5 Orientation

The determinant sign records orientation:

$$\det A > 0$$

preserves orientation, while

$$\det A < 0$$

reverses orientation.

The scalar triple product gives signed volume:

$$\det \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} = (u \times v) \cdot w.$$

If this determinant is zero, the three vectors lie in the same plane.

6 Common Mistakes

Mistake	Correct exam habit
Using Cramer's rule when $\det A = 0$.	First check invertibility.
Forgetting the transpose in the inverse formula.	Use $A^{-1} = C^T / \det A$.
Dropping the sign of area too early.	Use signed area for orientation; absolute value for size.
Writing $u \times v = v \times u$.	Cross product changes sign: $v \times u = -(u \times v)$.
Using cross product outside $\mathbb{R}^3$.	The standard cross product is a 3D tool.

7 Final Exam Checklist

Checklist.

1. Check whether $\det A \neq 0$ before using inverse formulas.
2. Build the cofactor matrix C .
3. Use $A^{-1} = C^T / \det A$.
4. For Cramer's rule, replace one column at a time by b .
5. Use determinant sign for orientation.
6. Use absolute determinant for area or volume size.
7. Use $u \times v$ for a normal vector in $\mathbb{R}^3$.
8. Check $u \cdot (u \times v) = 0$ and $v \cdot (u \times v) = 0$.